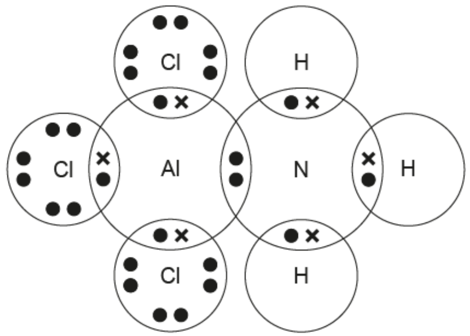
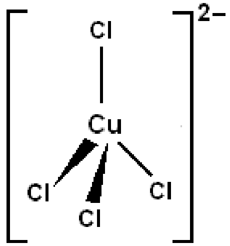
Section A

Question	Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
1	$3d^7$ allow any clear representation		1		1		
2	 NH_3 structure (1) dots and crosses showing coordinate bond and covalent bonds around Al atom (1)		2		2		
3	$K_w = [H^+][OH^-]$	1			1		
4	platinum as it is unreactive	1			1		
5	$SiCl_4$ has available d-orbitals whilst carbon does not (so water can bond to Si atom)	1			1		
6	forms a gas and gases have a greater entropy than liquids (and solids)		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	*(a)*			$Fe_3O_4 + 4CO \rightarrow 3Fe + 4CO_2$ ignore state symbols		1		1		
	(b)			award (1) for either of following • stability of +2 oxidation state increases down the group (due to the inert pair effect) • carbon stable as +4 and lead stable as +2	1			1		
8						1		1		1
				Section A total	**4**	**6**	**0**	**10**	**0**	**1**